

AIM: Perform a group-by operation in Spark DataFrames to compute the average of each group.

CODE:

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import avg

print("S091 Shreyash Kadam")

# Create Spark session
spark = SparkSession.builder \
    .appName("GroupByAverage") \
    .master("local[*]") \
    .getOrCreate()

# Read the CSV file
df = spark.read.csv(
    "students.csv",
    header=True,
    inferSchema=True
)

# Display the original data
print("Original Data:")
df.show()

# Group by Department and calculate average marks
result = df.groupBy("Subject").agg(
    avg("Marks").alias("Average_Marks")
)

# Display the result
print("Average Marks by Department:")
result.show()

# Stop Spark
spark.stop()
```

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OUTPUT:

```
"C:\Users\Asus\OneDrive\Documents\Apps\Attachments\Desktop\Shreyash SYCS SEM 3\Scala\MODULE 2\Practical M2_9_10\.venv\Scripts\python.exe" "C:\Users\Asus\OneDrive\Documents\Apps\Attachments\Desktop\Shreyash SYCS SEM 3\Scala\
S091 Shreyash Kadam
WARNING: Using incubator modules: jdk.incubator.vector
26/10/03 22:33:27 WARN Shell: did not find winutils.exe: java.io.FileNotFoundException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir are unset. -see https://wiki.apache.org/confluence/display/HADOOP2/Win
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
C:\Users\Asus\OneDrive\Documents\Apps\Attachments\Desktop\Shreyash SYCS SEM 3\Scala\MODULE 2\Practical M2_9_10\.venv\Lib\site-packages\pyspark\testing\utils.py:127: FutureWarning: PySpark does not yet fully support pandas >
require_minimum_pandas_version()
26/10/03 22:33:45 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Original Data:
-----+-----+
| Name|      Subject|Marks|
-----+-----+
|Shreyash|      Scala| 72|
|Om|      Python| 85|
|Krish|Operating System| 65|
|Palak|      Python| 91|
|deepak|Operating System| 78|
|Krishna|      Scala| 82|
|Abin|      Scala| 98|
|selvesh|      Python| 90|
|sahil|      Operat| 70|
-----+-----+

Average Marks by Department:
-----+-----+
|      Subject|      Average_Marks|
-----+-----+
|Operating System|      71.5|
|      Operat|      70.0|
|      Scala|      84.0|
|      Python|88.66666666666667|
-----+-----+

Process finished with exit code 0
```

AIM: Join two CSV files in Spark DataFrames based on a common column and write the output to a file.

CODE:

```
from pyspark.sql import SparkSession
```

```
spark = SparkSession.builder \
    .appName("JoinCSVFiles") \
    .master("local[*]") \
    .getOrCreate()
```

```
# Read stock CSV
```

```
stock_data = spark.read.csv(
    "stocks.csv",
    header=True,
    inferSchema=True
)
```

```
# Read company information CSV
```

```
company_data = spark.read.csv(
    "company_info.csv",
    header=True,
    inferSchema=True
)
```

```
# Join using common column
```

```
joined_data = stock_data.join(company_data, "symbol")
```

```
print("\nJoined Data:")
```

```
joined_data.show()
```

```
# Write joined data to CSV file
```

```
output_path = "joined_output.csv"
```

```
joined_data.toPandas().to_csv(output_path, index=False)
```

```
print("\nJoined data written successfully to joined_output.csv")
```

```
spark.stop()
```

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OUTPUT:

```
"C:\Users\Asus\OneDrive\Documents\Apps\Attachments\Desktop\Shreyash SYCS SEM 3\Scala\MODULE 2\Practical M2_9_10\.venv\Scripts\python.exe" "C:\Users\Asus\OneDrive\Documents\Apps\
WARNING: Using incubator modules: jdk.incubator.vector
26/10/03 22:38:59 WARN Shell: Did not find winutils.exe: java.io.FileNotFoundException: java.io.FileNotFoundException: HADOOP_HOME and hadoop.home.dir are unset. -see https://cwiki.apache.org/contrib/hadoop.html#howto\_config\_hadoop\_for\_windows
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
C:\Users\Asus\OneDrive\Documents\Apps\Attachments\Desktop\Shreyash SYCS SEM 3\Scala\MODULE 2\Practical M2_9_10\.venv\Lib\site-packages\pyspark\testing\utils.py:127: FutureWarning:
    require_minimum_pandas_version()
26/10/03 22:39:00 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

Joined Data:
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|symbol|   date| open|high| low|close| volume|   company| sector|country|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
|  AAL|2017-01-03| 44.5|46.2|44.1| 45.8|1250000|American Airlines|Airlines|  USA|
|  AAL|2017-01-04| 45.9|47.1|45.5|46.75|1320000|American Airlines|Airlines|  USA|
|  AAL|2017-01-05| 46.8|48.2|46.3| 47.9|1450000|American Airlines|Airlines|  USA|
|  AAL|2017-01-06|47.95|49.0|47.5| 48.6|1380000|American Airlines|Airlines|  USA|
|  AAL|2017-01-09| 48.7|50.2|48.2|49.85|1510000|American Airlines|Airlines|  USA|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+

C:\Users\Asus\OneDrive\Documents\Apps\Attachments\Desktop\Shreyash SYCS SEM 3\Scala\MODULE 2\Practical M2_9_10\.venv\Lib\site-packages\pyspark\sql\pandas\conversion.py:298: Futu
    require_minimum_pandas_version()
C:\Users\Asus\OneDrive\Documents\Apps\Attachments\Desktop\Shreyash SYCS SEM 3\Scala\MODULE 2\Practical M2_9_10\.venv\Lib\site-packages\pyspark\sql\pandas\conversion.py:348: User
    [PACKAGE_NOT_INSTALLED] PyArrow >= 18.0.0 must be installed; however, it was not found.
Attempting non-optimization as 'spark.sql.execution.arrow.pyspark.fallback.enabled' is set to true.
    warn(msg)

Joined data written successfully to joined_output.csv

Process finished with exit code 0
```